

Problem 5: Alien Protein Codes

(13 marks)

Dr. Muzabique figured out how to manipulate six alien proteins to do three functions: construct, cut, and pack. The alien genetic code used to manipulate the proteins was found to have 6 “alphabet letters”: A, T, G, C, D, N. Only he knew the top-secret algorithm, and two weeks ago, he was mysteriously found dead in his apartment.

The top-secret project is now in a fund crunch but the government decides to give it one more chance if the algorithm can be decoded. They send you the only page that was recovered from Dr. Muzabique’s notebook, with one set of input instructions and output genetic codes, and ask you to decode the algorithm:

“Today was a great breakthrough! I found that large and small proteins inherently behave in different ways. The way they operate under different cellular functions...”

Input: Celebi, construct Articulo

Output: CNDATACACCNDTGTATTGCGGGATGCNDTGCCACCCCNNDCTTTTCCACTNANCND

Input: Articulo, construct and cut Terrakion

Output: CNDTGTATTGCGGGATGCNDACTGCAGGGACTCCNDGCATGACATCNNNNNCNDTGCCACCC
CNND

Input: Terrakion, construct and pack Shaymin

Output: CNDACTGCAGGGACTCCNDCCAGGCTCNDGCATGACATCNNNNNCNDCTTTTCCACTNNNCND

Input: Cressela, construct and cut and pack Azelf

Output: CNDCCATATGATGCGACNDGTTGATCNDGCATGACATCNNNNNCNDTGCCACCCCNANCND
CTTTTCCACTNNNCND

Input: Celebi, cut Cressela

Output: CNDATACACCNDCCATATGATGCGACNDGCATGACATCANNNNNCNDCTTTTCCACTNANCND

Input: Shaymin, pack and construct Azelf

Output: CNDCCAGGCTCNDGTTGATCNDTGCCACCCCNANCND

Input: Celebi, construct and cut and pack Cressela

Output: CNDATACACCNDCCATATGATGCGACND

Input: Terrakion, pack Articulo

Output: CNDACTGCAGGGACTCCNDTGTATTGCGGGATGCNDCTTTTCCACTNNNCND

Assignment 1: (13 marks)

Preserve the legacy of Dr. Muzabique! Explain how the algorithm converts input commands into output genetic codes.

– Anshuman Swain